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DEVELOPING A REAL-TIME 3D TELEMETRY VISUALIZATION FOR STAIRLIFT FAULT DIAGNOSIS

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Abstract. This research addresses the challenge of transforming raw stairlift telemetry data into an intuitive 3D visualization for fault diagnosis. The core technical problem is the real-time mapping of noisy encoder, spindle, and swivel sensor readings onto a curved 3D rail geometry, requiring state estimation, spline interpolation, and quaternion-based rotation. A "virtual ghost" prototype was developed iteratively and validated through calibration runs on physical hardware. The results show that a pipeline combining Kalman filtering, piecewise encoder mapping, and Catmull-Rom spline interpolation produces a stable, real-time 3D representation that enables faster interpretation of stairlift position and fault conditions compared to text-based error codes.

1 Introduction

1.1 Context

DeVi Comfort designs and manufactures stairlift systems and develops the accompanying control software in-house. A stairlift consists of a chair unit that travels along a custom-bent rail mounted to a staircase. The chair is driven by a traction motor, while a spindle motor adjusts seat tilt and a swivel motor rotates the seat at landing positions. During operation, embedded controllers continuously produce telemetry: traction encoder pulses, spindle encoder values, swivel angles, and system status bits including error codes.

For service, maintenance, and further development, engineers need to quickly understand the current state of a stairlift: where the chair is positioned on the rail, whether it is moving correctly, and what deviations or faults have occurred.

1.2 Problem

Fault information from the stairlift controller is currently presented as textual error codes and numeric status values. While technically correct, these representations require significant domain knowledge to interpret spatially. An error code indicating a position deviation, for example, does not immediately convey where on the rail the deviation occurred, in which direction, or how large the discrepancy is.

The underlying technical challenge is twofold. First, the raw telemetry data (encoder pulse counts, spindle values, swivel angles) must be mapped onto a three-dimensional rail geometry that contains straight sections, vertical bends, and horizontal bends, each with different encoder pulse costs per unit of travel. Second, the sensor readings arrive at irregular intervals over a wireless serial link with noise, latency, and occasional packet loss. The visualization must bridge the gap between these asynchronous, noisy measurements and a continuous, smooth spatial representation that updates at display frame rate.

1.3 Relevance

A reliable 3D visualization of stairlift position and movement has direct practical value. Service engineers can visually identify whether the chair has lost its expected position, is stuck in a bend, or exhibits abnormal tilt. This reduces diagnosis time and minimizes unnecessary on-site visits. For the development team, a live telemetry view enables rapid verification of firmware behavior during testing and commissioning of new rail configurations.

1.4 Research Goal

The goal of this research is to determine which data processing and mathematical models are needed to transform live stairlift telemetry into a reliable and intuitive real-time 3D visualization for fault analysis.

The central research question is:

What data processing pipeline and mathematical models are required to transform live stairlift telemetry into a real-time 3D visualization that enables faster and more intuitive fault diagnosis compared to text-based error codes?

1.5 Approach

The research follows an iterative prototyping approach based on the spiral model [2]. Each iteration consists of analysis, design, implementation, and validation against practical scenarios on physical hardware. The primary methods employed are formal

mathematical analysis, architectural design, and experiment with measurable accuracy criteria.

1.6 Reading Guide

Section 2 presents the problem statement, structured as a problem analysis, prior work survey, and the resulting research questions. Section 3 describes the research methods per subquestion. Section 4 presents the findings. Section 5 concludes with answers to the research questions and recommendations.

2 Problem Statement

2.1 Problem Analysis

2.1.1 Sensor-to-Spatial Mapping in Mechatronic Systems

The challenge of transforming discrete sensor readings into a continuous spatial representation is well-established in mechatronic system diagnostics [8]. In the broader context of digital twin research, Bofill et al. identify real-time sensor data integration as one of the primary technical barriers to effective fault monitoring [3]. The fundamental difficulty is that sensors produce scalar values (counts, voltages, status bits) at discrete intervals, while a spatial visualization requires a continuous 3D position and orientation that updates at display frame rate.

In robotics and vehicle tracking, this problem is typically simplified by the assumption that sensor readings map linearly to spatial coordinates [6]. A rotary encoder on a linear actuator, for example, produces pulses proportional to displacement, and the mapping from pulse count to position is a single constant. This assumption breaks down when the trajectory is non-linear: on a curved path, the relationship between encoder pulses and spatial displacement depends on the local curvature [14].

2.1.2 Non-Linear Encoder Mapping on Curved Rails

A stairlift rail is composed of straight sections, vertical bends (pitch changes), and horizontal bends (yaw changes). The mechanical drive system uses a traction motor with an incremental encoder to measure displacement [4]. Due to the geometry of the rail bending mechanism, each segment type consumes a different number of encoder pulses per unit of physical travel. A vertical upward bend, for instance, requires fewer pulses per millimeter of arc length than a downward bend, because the traction wheel follows the inner or outer radius of the curve respectively.

This means that a single scalar encoder reading cannot be converted to a 3D position without a model of the full rail topology. The mapping must account for the cumulative pulse cost of each preceding segment, making it a piecewise non-linear function rather than a simple linear scaling. Without such a model, position errors accumulate at every bend transition, growing proportionally to the number of bends in the rail.

2.1.3 Real-Time State Estimation from Noisy Telemetry

The telemetry link between the stairlift controller and the visualization application introduces additional challenges. The Bluetooth Classic Serial Port Profile [1] provides a virtual serial connection at approximately 20 Hz, but with variable latency (10–80 ms per round trip) and occasional packet loss. The visualization, however, must render at 60 fps (16.7 ms per frame).

This creates a temporal mismatch: new sensor data arrives roughly every third frame, and frames between measurements must be filled with predicted values. If the most recent measurement is simply held constant until the next arrives, the display exhibits visible staircase-like jumps. If measurements are used directly without filtering, encoder quantization noise (inherent to incremental encoders [14]) causes jitter. Both artifacts undermine the interpretability of the visualization.

The general solution to this class of problem is recursive state estimation: a filter that maintains a probabilistic model of the system state and updates it with each new measurement [9]. The choice of filter, its process model, and its tuning determine the trade-off between responsiveness and smoothness.

2.1.4 Multi-Axis Orientation from Independent Sensors

The stairlift chair has four independent rotational degrees of freedom: yaw (direction along the rail), pitch (rail inclination), seat tilt (spindle motor), and seat rotation (swivel motor). Each is measured by a separate sensor with its own update rate and noise characteristics. Reconstructing the full 3D orientation of the chair requires combining these independent streams into a single consistent rotation, avoiding mathematical singularities that can occur when multiple rotations are composed [15].

2.1.5 Summary of Technical Subproblems

Three interrelated technical subproblems emerge from this analysis:

1. **Non-linear encoder-to-position mapping:** Converting a scalar encoder count into a normalized position along a 3D curve with segment-dependent pulse costs.

2. **Real-time state estimation:** Producing a smooth, low-latency estimate of position and velocity from noisy, asynchronous measurements arriving at a lower rate than the display refresh.
3. **Multi-axis orientation reconstruction:** Combining four independent rotational sensor streams into a singularity-free 3D orientation.

2.2 Prior Work

2.2.1 State Estimation

For real-time state estimation from noisy sensor data, the Kalman filter is the standard approach in control engineering and robotics [9, 16]. The filter maintains a probabilistic estimate of system state (position and velocity) and updates it with each new measurement, optimally balancing prediction uncertainty against measurement noise [4]. Alternatives include simple moving average filters and complementary filters. Moving averages introduce latency proportional to the window size and cannot predict between measurements. Complementary filters combine two sensor sources (e.g., accelerometer and gyroscope) but assume complementary frequency characteristics, which does not apply to a single-encoder system. The Kalman filter is therefore the most suitable choice for this application: it provides both filtering and inter-measurement prediction in a single framework.

2.2.2 Spline Interpolation for Smooth Trajectories

To produce smooth position and orientation transitions along a discrete set of rail control points, interpolation is required. Linear interpolation produces sharp direction changes at segment boundaries. Cubic B-splines offer smoothness but do not pass through the control points, complicating correspondence with physical rail geometry. Catmull-Rom splines pass through each control point and provide C^1 continuity (smooth tangent transitions), making them a natural choice for trajectory following along a known set of waypoints [5].

2.2.3 Rotation Representation

Representing 3D orientation requires choosing between Euler angles, rotation matrices, and quaternions. Euler angles suffer from gimbal lock at certain configurations and require careful ordering of rotation axes [15]. Rotation matrices are gimbal-lock-free but costly to interpolate. Quaternions provide compact, gimbal-lock-free representation with efficient interpolation (SLERP) and are the standard in real-time 3D applications [10].

2.2.4 Existing Work at DeVi Comfort

Prior to this research, fault information within DeVi Comfort was presented exclusively as text-based error codes and numeric register values. No spatial visualization of stairlift telemetry existed. The IPC (Inter-Process Communication) infrastructure connecting the configurator application to the lift controller was limited to rail upload and download; live telemetry streaming had not been implemented.

2.3 Research Questions

The problem analysis identified three technical subproblems (non-linear encoder mapping, real-time state estimation, and multi-axis orientation reconstruction), while the prior work survey identified candidate approaches for each. The following subquestions decompose the central research question into independently answerable parts, each motivated by a specific gap between the identified problem and the available solutions.

1. *Which fault types and status indicators from the stairlift controller are most relevant for spatial visualization?* The firmware produces over 168 message types. Before any visualization can be designed, the subset that carries spatially interpretable information must be identified and classified.
2. *What data pipeline architecture can transport live telemetry from the controller to a 3D rendering environment at sufficient throughput?* The problem analysis showed that the Bluetooth serial link introduces latency, noise, and packet loss. The pipeline must handle these at each layer while maintaining the $\sim$20 Hz update rate.
3. *How can encoder pulse counts be mapped to a normalized position along a 3D rail with segment-dependent pulse costs?* As established in Section 2.1, the non-linear relationship between encoder pulses and physical distance is the core geometric challenge. No existing mapping exists for this rail type.
4. *Which mathematical models for state estimation, interpolation, and rotation produce a stable real-time 3D representation from noisy telemetry?* The prior work survey identified several candidate models per subproblem. This question addresses the formal derivation and parameterization of the selected models for this specific application.
5. *Which rendering techniques make position deviations and fault conditions most interpretable in a 3D environment?* The visualization must convey multiple simultaneous data streams (position, tilt, rotation, accessory states) without overwhelming the operator. The choice of visual encoding affects diagnostic effectiveness.
6. *How can the prototype be validated against practical scenarios, and what measurable criteria determine sufficient accuracy?* The mathematical models must be verified under real-world conditions where mechanical tolerances, Bluetooth variability, and sen-

sor noise interact.

3 Methods

Each subquestion is addressed with one or more research methods. This section describes, per subquestion, which method is used, why it is appropriate, how it is concretely applied, and what criteria determine completion.

3.1 Fault Relevance (Observation Study)

Method: Observation study of the existing telemetry protocol.
Justification: The stairlift controller firmware defines over 168 message types. To determine which are most relevant for visualization, the existing protocol documentation and firmware behavior must be systematically catalogued.
Approach: The binary protocol specification is reverse-engineered from the firmware message definitions. Each telemetry message type is classified by content (position, speed, angle, status, error) and polling frequency. Relevance is assessed based on spatial interpretability and diagnostic value.
Instrument: A classification table listing each message type, its data fields, update rate, and relevance score for spatial visualization.

3.2 Data Pipeline Architecture (Design)

Method: Architectural design.
Justification: The telemetry pipeline spans multiple layers (Bluetooth transport, binary protocol parsing, coroutine-based polling, signal dispatch) and must satisfy real-time constraints. A structured architectural design ensures that each layer is independently testable and that throughput requirements are met.
Approach: The architecture is designed as a layered pipeline: transport abstraction, protocol parser, telemetry poller, state estimator, and 3D renderer. Each layer is specified with its inputs, outputs, timing constraints, and failure modes.
Instrument: A layered architecture diagram with throughput and latency specifications per layer.

3.3 Encoder-to-Position Mapping (Formal Analysis)

Method: Formal mathematical analysis.
Justification: The encoder-to-position mapping is a deterministic geometric problem: given a rail composed of segments with known curvature and known pulse costs per segment type, the mapping can be derived analytically. A formal derivation ensures correctness and makes the mapping reproducible.
Approach: The rail is modeled as a sequence of segments (straight, vertical bend, horizontal bend). For each segment type, the encoder pulse cost is derived from firmware constants. Cumulative 3D Euclidean distances are computed at segment boundaries, creating a piecewise linear mapping from encoder value to normalized rail position. The derivation includes the arc-tracing geometry used to construct 3D control points from the firmware's segment description.
Instrument: The formal derivation is considered complete when: (a) the mapping is defined for all segment types, (b) a calibration procedure aligns encoder values with 3D distances, and (c) the mapping error is below 5 mm on a reference rail.

3.4 Mathematical Models (Formal Analysis)

Method: Formal mathematical analysis.
Justification: The state estimation, interpolation, and rotation models are grounded in established mathematical theory (Kalman filtering, Catmull-Rom splines, quaternion algebra). Formal derivation from first principles ensures that the models are correctly applied and that tuning parameters have a clear theoretical basis.
Approach: Three models are derived:

- A 2D constant-velocity Kalman filter for each telemetry axis (traction, spindle, swivel), including the process noise matrix derivation, measurement update equations, and inter-measurement dead-reckoning extrapolation.
- Catmull-Rom spline interpolation for tangent computation along the rail, paired with linear position interpolation on segments.
- Quaternion-based orientation reconstruction for pitch (from rail geometry), spindle compensation, and swivel rotation.

Instrument: Each model is presented as a complete mathematical derivation with named equations. Correctness is verified by comparing predicted positions against known reference positions on a test rail.

3.5 Rendering Techniques (Design + Experiment)

Method: Design combined with informal comparative evaluation.
Justification: The choice of rendering technique affects interpretability. Multiple approaches are prototyped and compared based on visual clarity and information density.
Approach: Several visualization techniques are implemented: a ghost chair model following the rail, semi-transparent ghost stairs for spatial context, a dither-clip effect for depth perception, and rail annotations for segment identification. Each technique is evaluated in context of a running stairlift scenario.
Instrument: A comparison matrix listing each technique, its purpose, visual characteristics, and suitability for fault identification.

3.6 Validation (Experiment)

Method: Experiment with measurable criteria.
Justification: The prototype must be validated on physical hardware to verify that the mathematical models and pipeline architecture produce correct results under real-world conditions (Bluetooth latency, mechanical tolerances, sensor noise).
Approach: A calibration run is executed on a physical stairlift installation. The lift is driven from bottom to top while the prototype records encoder positions, applies the mathematical models, and renders the 3D ghost in real time. At horizontal bends, compass-based angle corrections are measured and compared against expected values.
Instrument: The experiment is evaluated on: (a) positional accuracy (deviation between displayed and actual position at known reference points), (b) angle correction accuracy at horizontal bends, (c) visual smoothness (absence of jitter at 60 fps), and (d) end-to-end latency from sensor reading to display update.

4 Results

4.1 Fault Types and Telemetry Classification

Analysis of the stairlift firmware protocol reveals 168 message types, of which six are directly relevant for real-time spatial visualization. These are classified in Table 1.

The three high-frequency streams (traction, spindle, swivel) form the core input for the 3D visualization. They are polled in an alternating pattern to share the serial Bluetooth link: the fast loop alternates between traction and spindle every frame tick, while a parallel coroutine polls swivel at 50 ms intervals. The low-frequency streams (system status, footrest) update chair accessory states every 2 seconds.

Status bits relevant for fault visualization include: armrest position (left and right, from `data[14]` bits 0 and 1 of MSG 114), seatbelt state (bit 2), and footrest position (bit 2 of MSG 113, inverted logic). These are mapped directly to accessory animations on the 3D chair model.

4.2 Data Pipeline Architecture

The telemetry pipeline is structured as five layers, shown in Figure 1.

Transport layer. Abstracts Bluetooth Classic and TCP connections behind a common interface. The Bluetooth transport communicates with the HC-06 module on the lift controller. A 40-byte null flush clears the module's receive buffer before each session. Message counters increment from 1 to 200 and wrap around.

Protocol parser. A six-state finite state machine [7] processes the byte stream: `WAIT_START` → `MSG_NUMBER` → `COMMAND_ID` → `LENGTH` → `DATA` → `CHECKSUM`. The wire format is:

$$[\texttt{0x0A}][\text{MsgNr}][\text{RwId}][\text{CmdId}][\text{Len}][\text{Data}_{0..39}][\text{Chk}] \tag{1}$$

where the checksum is computed as:

$$\text{Chk} = (\texttt{0x0A} + \text{MsgNr} + \text{RwId} + \text{CmdId} + \text{Len} + \textstyle\sum \text{Data}_i) \bmod 256 \tag{2}$$

Invalid lengths (> 40) or checksum mismatches trigger a resync to `WAIT_START`.

Telemetry poller. Three concurrent coroutine loops share the serial link cooperatively:

- *Fast loop* (∼20 Hz): alternates traction (MSG 110) and spindle (MSG 111) requests each frame tick.
- *Swivel loop* (∼20 Hz): polls the swivel angle via the APP_TO_BUS multiplexed channel (MSG 168) at 50 ms intervals.
- *Slow loop* (0.5 Hz): polls system status (MSG 114) and footrest state (MSG 113) every 2 s.

Data parsing. The traction encoder is extracted as a 32-bit big-endian integer from bytes 1–4 of the MSG 110 response. The spindle encoder is a 16-bit value from bytes 1–2 of MSG 111. The swivel angle is computed from MSG 168 as:

$$\theta_{\text{swivel}} = \frac{(\texttt{data[3]} \ll 8 \mid \texttt{data[4]}) - \texttt{0x7FFF}}{18} \tag{3}$$

4.3 Encoder-to-Position Mapping

4.3.1 Rail Geometry Construction

The lift controller stores the rail as a sequence of typed blocks: straight sections, vertical bends (up/down), and horizontal bends (left/right), each measured via incremental encoders [4]. Each block specifies a type and an amount. To construct a 3D representation, these blocks are traced sequentially using arc geometry. Figure 2 shows the three segment types and their arc-tracing geometry.

PLACEHOLDER: rail_segment_types.png

Figure 2: The three rail segment types: straight (left), vertical bend (center), and horizontal bend (right), with their respective arc-tracing geometry and encoder pulse costs.

Table 1: Telemetry message types relevant for spatial visualization.

Message	Data	Rate	Use
MSG 110	Traction encoder (32-bit)	20 Hz	Rail position
MSG 111	Spindle encoder (16-bit)	20 Hz	Seat tilt angle
MSG 168	Swivel encoder (16-bit)	20 Hz	Seat rotation
MSG 114	System status bits	0.5 Hz	Arm/seatbelt state
MSG 113	Footrest status	0.5 Hz	Footrest position
MSG 44	Pilot state	On demand	Calibration run

```
+-----------------+    +-----------------+    +-----------------+    +-----------------+    +-----------------+
| Transport       | -> | Protocol Parser | -> | Telemetry Poller| -> | State Estimator | -> | 3D Renderer     |
| (Bluetooth/TCP) |    | (6-state FSM,   |    | (3 coroutines,  |    | (3x Kalman      |    | (Ghost Chair    |
|                 |    |  checksum)      |    |  20 Hz polling) |    |  Filter)        |    |  + Ghost Stairs)|
+-----------------+    +-----------------+    +-----------------+    +-----------------+    +-----------------+
     raw bytes          LiftMessage objs        typed signals        filtered pos/angle      60 fps display
```

Figure 1: Layered telemetry pipeline architecture from Bluetooth transport to 3D rendering.

The direction vector at any point along the rail is determined by the current yaw (ψ) and pitch (ϕ):

$$\mathbf{d}(\psi, \phi) = \begin{pmatrix} -\sin\psi\cos\phi \\ \sin\phi \\ -\cos\psi\cos\phi \end{pmatrix} \tag{4}$$

For straight sections, the position advances by the section length along $\mathbf{d}$. For bends, each segment subtends a fixed angle $\Delta\alpha = 5^\circ$ at a bend radius $r = 150\,\text{mm}$. The arc length per segment is:

$$s = r \cdot \Delta\alpha_{\text{rad}} = 0.150 \cdot \frac{5\pi}{180} \approx 0.01309\,\text{m} \tag{5}$$

Position is updated using a trapezoidal midpoint approximation:

$$\mathbf{p}_{i+1} = \mathbf{p}_i + \frac{\mathbf{d}_{\text{before}} + \mathbf{d}_{\text{after}}}{2} \cdot \frac{s}{\left\| \frac{\mathbf{d}_{\text{before}} + \mathbf{d}_{\text{after}}}{2} \right\|} \tag{6}$$

where $\mathbf{d}_{\text{before}}$ and $\mathbf{d}_{\text{after}}$ are the direction vectors before and after the pitch or yaw increment.

4.3.2 Piecewise Encoder Mapping

Different rail segment types consume different numbers of encoder pulses per unit of travel. The firmware defines the following constants:

$$\begin{aligned} k_{\text{straight}} &= 3.361352\,\text{pulses/mm} \\ k_{\text{up}} &= 37.547\,\text{pulses/segment} \\ k_{\text{down}} &= 50.453\,\text{pulses/segment} \\ k_{\text{horiz}} &= 44.0\,\text{pulses/segment} \\ E_{\text{start}} &= 100\,\text{pulses (initial offset)} \end{aligned} \tag{7}$$

A piecewise linear mapping is constructed by computing, at each segment boundary, both the cumulative encoder value and the cumulative 3D Euclidean distance from the rail start. For an encoder reading E, the normalized position $t \in [0, 1]$ is found by:

1. Locating the segment $[E_i, E_{i+1}]$ containing E.
2. Linearly interpolating the 3D distance: $d = d_i + (d_{i+1} - d_i) \cdot \frac{E - E_i}{E_{i+1} - E_i}$
3. Normalizing: $t = d/d_{\text{total}}$

This approach accounts for the non-linear relationship between encoder pulses and physical distance that arises at bends.

4.3.3 Spindle Angle Lookup

The spindle encoder value is converted to a seat tilt angle using a 751-entry lookup table ported from the firmware's `Curve.h`. The table maps angles from 0.0° to 75.0° in 0.1° increments to corresponding encoder pulse counts. Given a raw encoder reading, a base offset of 10 000 is subtracted, and an 11-iteration binary search locates the corresponding angle:

$$\theta_{\text{spindle}} = \frac{\text{bisect}(\text{TABLE}, \max(0, E_{\text{raw}} - 10000))}{10} \tag{8}$$

4.4 Mathematical Models for State Estimation

4.4.1 Kalman Filter

A 2D constant-velocity Kalman filter [9, 16] is applied independently to each telemetry axis (traction position, spindle angle, swivel angle). The state vector is $\mathbf{x} = [p, v]^T$ (position and velocity).

Prediction step (time update with interval Δt):

$$\begin{aligned} p &\leftarrow p + v \cdot \Delta t \\ P_{00} &\leftarrow P_{00} + 2\Delta t \cdot P_{01} + \Delta t^2 \cdot P_{11} + \sigma_a^2 \cdot \tfrac{\Delta t^3}{3} \\ P_{01} &\leftarrow P_{01} + \Delta t \cdot P_{11} + \sigma_a^2 \cdot \tfrac{\Delta t^2}{2} \\ P_{11} &\leftarrow P_{11} + \sigma_a^2 \cdot \Delta t \end{aligned} \tag{9}$$

where σ_a is the process noise (acceleration standard deviation) and $\mathbf{P}$ is the covariance matrix. The process noise matrix corresponds to the continuous

white-noise-jerk model:

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^2/2 & \Delta t \end{bmatrix} \tag{10}$$

Measurement update (correction with measurement z):

$$\begin{aligned} y &= z - p \\ S &= P_{00} + \sigma_m^2 \\ K_0 &= P_{00}/S \\ K_1 &= P_{01}/S \\ p &\leftarrow p + K_0 \cdot y \\ v &\leftarrow v + K_1 \cdot y \end{aligned} \tag{11}$$

where σ_m is the measurement noise standard deviation and $\mathbf{K}$ is the Kalman gain.

The tuning parameters per axis are listed in Table 2.

4.4.2 Dead-Reckoning Extrapolation

Between measurement updates, the display position is extrapolated from the Kalman filter state:

$$p_{\text{target}} = p_{\text{KF}} + v_{\text{KF}} \cdot t_{\text{elapsed}} \tag{12}$$

After 0.8 s without a measurement, velocity decays exponentially:

$$v \leftarrow v \cdot e^{-5.0 \cdot (t_{\text{elapsed}} - 0.8)} \tag{13}$$

The display position is then smoothed with a frame-rate-independent exponential filter [11]:

$$p_{\text{display}} = \text{lerp}(p_{\text{display}},\ p_{\text{target}},\ 1 - e^{-\alpha \cdot \Delta t}) \tag{14}$$

where α is the smoothing rate from Table 2.

4.4.3 Catmull-Rom Spline Tangent

To determine the chair's orientation along the rail, the tangent direction at the current position is computed using a Catmull-Rom spline [5] through the surrounding control points $\mathbf{p}_0 \ldots \mathbf{p}_3$:

$$\mathbf{T}(t) = \tfrac{1}{2}\left[(-\mathbf{p}_0 + \mathbf{p}_2) + 2(2\mathbf{p}_0 - 5\mathbf{p}_1 + 4\mathbf{p}_2 - \mathbf{p}_3)t + 3(-\mathbf{p}_0 + 3\mathbf{p}_1 - 3\mathbf{p}_2 + \mathbf{p}_3)t^2\right] \tag{15}$$

The position along segments uses linear interpolation rather than the full Catmull-Rom curve, ensuring the chair stays on the rail segments while rotating smoothly through transitions.

4.4.4 Quaternion Orientation

The chair yaw is computed from the tangent's horizontal projection [10]:

$$\psi = \text{atan2}(T_x, T_z) \tag{16}$$

with phase unwrapping to avoid 2π discontinuities.

Rail chirality (which side the chair faces) is determined by the cumulative cross product of consecutive direction vectors projected onto the xz-plane:

$$C = \sum_i (d_x^i \cdot d_z^{i+1} - d_z^i \cdot d_x^{i+1}) \tag{17}$$

If $C \geq 0$, the rail is left-turning and the chair faces outward at $+90°$; otherwise at $+270°$.

The pitch tilt of the chair unit is applied via a world-to-local quaternion transform:

$$q_{\text{unit}} = q_{\text{parent}}^{-1} \cdot q(\mathbf{a}_{\text{pitch}},\ -\phi) \cdot q_{\text{parent}} \tag{18}$$

where $\phi = \text{atan2}(T_y, \|\mathbf{T}_{xz}\|)$ and $\mathbf{a}_{\text{pitch}} = \hat{\mathbf{y}} \times \hat{\mathbf{T}}_{\text{flat}}$.

The spindle motor compensates the rail pitch to keep the seat level, then adds its own tilt:

$$q_{\text{spindle}} = q_{\text{parent}}^{-1} \cdot q(\mathbf{a},\ -\phi + \theta_{\text{spindle}}) \cdot q_{\text{parent}} \tag{19}$$

A geometry blend factor of 15% mixes the expected pitch angle (from rail geometry) into the Kalman-filtered spindle angle:

$$\theta_{\text{target}} = 0.85 \cdot \theta_{\text{KF}} + 0.15 \cdot \phi_{\text{rail}} \tag{20}$$

4.5 Rendering Techniques

Five visualization techniques were implemented in a real-time 3D engine [12] and evaluated for their contribution to fault interpretability:

1. **Ghost chair model:** A multi-pivot 3D model (unit, swivel, spindle, fold, frame) loaded from GLB parts, following the rail position in real time. The pivot hierarchy enables independent animation of each mechanical degree of freedom.
2. **Ghost stair renderer:** Semi-transparent stair treads generated from the rail height profile. Treads are placed every 180 mm of vertical rise, with even distribution between platform landings. This provides spatial context for the chair's position on the staircase.
3. **Dither-clip effect:** A shader-based clipping technique that gradually reveals or hides geometry based on distance from the chair, reducing visual clutter while maintaining context.
4. **Rail annotations:** Labels overlaid on the 3D rail identifying segment types, boundaries, and charging station positions.
5. **Liquid glass UI:** A frosted-glass shader design system for information panels, optimized for both desktop and tablet rendering.

Figure 3 shows the ghost chair visualization tracking the stairlift position in real time along a curved rail with ghost stairs.

The ghost chair model combined with ghost stairs proved most effective for spatial fault diagnosis: the chair's position on the staircase is immediately apparent, and deviations in tilt or position are visible as misalignment between the chair and the expected rail trajectory.

Table 2: Kalman filter tuning parameters per telemetry axis.

Parameter	Traction	Spindle	Swivel
σ_a (process noise)	0.03	8.0	10.0
σ_m (measurement noise)	0.0002	0.2	0.5
Display smoothing rate	8.0	12.0	12.0
Max velocity	$1.2v_{\max}$	$30°/s$	$40°/s$

Figure 3: The virtual ghost visualization: the 3D ghost chair follows the rail in real time based on live telemetry. Semi-transparent ghost stairs provide spatial context. Rail annotations identify segment boundaries along the rail.

4.6 Validation Results

The prototype was validated through calibration runs on physical stairlift installations. During a calibration run, the lift is driven from bottom to top (and back) while the system polls telemetry at 250 ms intervals. At each horizontal bend, a compass-based angle correction is computed and compared against the expected value.

PLACEHOLDER: calibration_run.png

Figure 4: The calibration run in progress: the ghost chair tracks the physical lift position while compass-based angle corrections are applied at horizontal bends.

Calibration run protocol. The service polls six data points per tick: lock state, traction encoder, pilot state (keep-alive), spindle encoder, lift location, and (every 4th tick) error status. Speed is managed automatically: 50 (normal) for straight sections and 20 (reduced) for bends and short straights under 250 mm.

Angle correction. At horizontal bends, the compass tracker measures accumulated yaw via gyroscope integration:

$$\psi_{\text{acc}} \mathrel{+}= \omega_z \cdot \frac{180}{\pi} \cdot \Delta t \tag{21}$$

The correction is computed as:

$$\delta = |\theta_{\text{target}}/2| - |\psi_{\text{measured}}| \tag{22}$$

where $\theta_{\text{target}} = n_{\text{segments}} \cdot 5°$ is the expected total bend angle. This correction is transmitted to the firmware as a scaled 16-bit value ($\text{round}(\delta \cdot 8.0)$).

Split-bend compensation. When two horizontal bends of the same direction are separated by a short straight section (< 250 mm), the first bend's accumulated yaw must be subtracted before measuring the second. The detector identifies this pattern and applies the correction automatically.

Quantitative results. Table 3 summarizes the measured performance across three calibration runs on different rail configurations.

Analysis of results. The position error at end stations remained below the 5 mm threshold defined in the

Table 3: Validation measurements across three calibration runs on physical stairlift installations.

Metric	Run 1 (6 bends)	Run 2 (4 bends)	Run 3 (8 bends)
Rail length	4 820 mm	3 210 mm	6 450 mm
Total encoder range	16 200 pulses	10 790 pulses	21 680 pulses
Position error at end station	< 3 mm	< 2 mm	< 5 mm
Max position error at bend transition	4.2 mm	2.8 mm	6.1 mm
Angle correction error (horizontal bends)	0.3° avg	0.5° avg	0.4° avg
Encoder jitter (raw, stationary)	± 2 pulses	± 3 pulses	± 2 pulses
Encoder jitter (after Kalman filter)	< 0.1 pulse	< 0.1 pulse	< 0.1 pulse
Display update latency (BT round trip)	35 ms avg	42 ms avg	38 ms avg
Render frame rate (desktop)	60 fps stable	60 fps stable	60 fps stable
Render frame rate (tablet, Android 12+)	55–60 fps	58–60 fps	52–58 fps

method (Section 3.3) across all three runs. The largest errors occurred at bend transitions on the longest rail (Run 3, 6.1 mm), which is explained by the accumulation of arc-tracing approximation errors over eight bends. The piecewise encoder mapping eliminated the position jumps at segment boundaries that were observed during early testing with a linear mapping.

The Kalman filter reduced stationary encoder jitter from ± 2–3 raw pulses to below 0.1 pulse equivalent in the display position, confirming that the chosen process noise parameters ($\sigma_a = 0.03$ for traction) provide sufficient smoothing without introducing perceptible lag. The display latency of 35–42 ms represents the Bluetooth round-trip time and is masked by the dead-reckoning extrapolation (Equation 12), which fills inter-measurement frames with predicted positions.

The spindle geometry blend (85% Kalman, 15% rail pitch) prevented the seat tilt oscillation that was observed during early testing with 100% Kalman filtering, where encoder noise caused visible vibration in the spindle angle.

On tablet hardware, frame rates occasionally dropped below 60 fps on the longest rail (Run 3) due to the higher polygon count of the ghost stair mesh. Shadow quality was automatically reduced on mobile to maintain acceptable performance [13].

5 Conclusion

5.1 Answers to Research Questions

Fault types. The classification of 168 firmware message types revealed that only six carry spatially interpretable information. This is a significant finding: it means that the bandwidth-constrained Bluetooth link does not need to carry the full telemetry stream. By targeting only these six messages with a multi-rate polling strategy, the system achieves 20 Hz position updates within the serial link's capacity. The implication is that future extensions (e.g., vibration or temperature monitoring) would require careful evaluation of whether the serial bandwidth can accommodate additional streams without degrading position update rates.

Pipeline architecture. The five-layer architecture proved effective in isolating failure modes: byte-level errors are caught by the protocol parser's checksum, while higher-level issues (connection loss, timeout) are handled by the transport layer. This separation was validated in practice when Bluetooth disconnections during calibration runs were recovered transparently without corrupting the state estimator. The cooperative coroutine approach to link sharing, while effective at 20 Hz, would not scale to significantly higher update rates without switching to a full-duplex transport.

Encoder mapping. The piecewise mapping resolved the core geometric problem identified in Section 2.1: the non-linear relationship between encoder pulses and physical distance at bends. The validation data (Table 3) confirms that position errors remain below 5 mm on rails up to 4.8 m, but increase to 6.1 mm on longer rails with more bends. This suggests that the trapezoidal arc-tracing approximation (Equation 6) introduces a small systematic error per bend that accumulates over the rail length. For rails with more than eight bends, a higher-order integration scheme may be needed.

Mathematical models. The constant-velocity filter effectively suppressed encoder jitter ($\pm$2–3 raw pulses reduced to $<$0.1 pulse equivalent) while maintaining responsiveness through dead-reckoning extrapolation. The tuning parameters (Table 2) were determined empirically; a formal sensitivity analysis would strengthen confidence in their generalizability to different rail configurations. The 85/15 blend between filtered spindle angle and expected rail pitch was necessary to prevent oscillation, but this ratio was found through trial and error rather than derived from first principles, which is a limitation of the current approach.

Rendering techniques. The ghost chair with ghost stairs provides immediate spatial context that text-based error codes fundamentally cannot offer. However, the current evaluation of rendering effectiveness is qualitative rather than quantitative. Without a controlled user study comparing diagnosis time and error rate, the claim of "faster and more intuitive" di-

agnosis remains an inductively supported hypothesis rather than a deductively proven conclusion.

Validation. The calibration runs confirm technical correctness of the mathematical models under real-world conditions. The position accuracy (<5 mm on standard rails), jitter suppression (factor >20 reduction), and frame rate (60 fps on desktop, 52–60 fps on tablet) meet or exceed the criteria defined in Section 3.6. The main limitation is sample size: three calibration runs on three rail configurations provide evidence of correctness but cannot establish statistical generalizability.

5.2 Answer to Central Question

The data processing pipeline required to transform live stairlift telemetry into a real-time 3D visualization consists of five layers: wireless transport, checksummed protocol parsing, multi-rate coroutine polling, recursive state estimation, and spline-interpolated 3D rendering. The mathematical foundation rests on three pillars: piecewise encoder mapping for non-linear rail geometries, a constant-velocity recursive filter for noise rejection and inter-measurement prediction, and quaternion-based orientation reconstruction from spline tangent vectors.

The validation data supports the deductive conclusion that, given a correct rail topology model and correctly tuned filter parameters, the pipeline produces positional accuracy within 5 mm and orientation accuracy within 0.5° at horizontal bends. The broader claim that this visualization enables faster fault diagnosis than text-based error codes is an inductive conclusion supported by the spatial immediacy of the 3D representation, but not yet quantified through a controlled user study.

5.3 Limitations

- The validation was performed on three rail configurations. Statistical generalizability to the full range of rail designs has not been established.
- The filter tuning parameters and spindle blend ratio were determined empirically. A formal sensitivity analysis is absent.
- The rendering effectiveness evaluation is qualitative. No controlled user study with measurable diagnosis time reduction has been conducted.
- The arc-tracing approximation introduces a small cumulative error per bend. On rails with many bends (>8), this may exceed the 5 mm threshold.

5.4 Recommendations

- **Controlled user study:** Measure diagnosis time and error rate with and without the 3D visualization on a standardized set of fault scenarios to convert the inductive claim into quantitative evidence.
- **Arc-tracing accuracy:** Replace the trapezoidal midpoint approximation with a higher-order numerical integration (e.g., Simpson's rule) to reduce cumulative bend error on long rails.
- **Filter tuning analysis:** Perform a sensitivity analysis on the process and measurement noise parameters to determine safe operating ranges and automate tuning per rail configuration.
- **Historical playback:** Record telemetry sessions for later replay to enable post-incident analysis without a live connection.

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